RIG-SMOOTH MORPHISMS ARE STABLE UNDER ADIC BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-base-change-rig-smooth; the proof environment immediately following it is its own proof

Editorial note. The label lemma-base-change-rig-smooth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the WAdm^Noeth-level base change lemma for rig-smooth arrows (lemma-base-change-rig-smooth-continuous of the same chapter)
- the Formal Spaces remark on the Noetherian variant of the base change comparison (formal-spaces-remark-base-change-variant-variant-Noetherian) and the adic morphism discussion (formal-spaces-section-adic)

Source reference. Original remark 0GBG: sources/stacks-chow/tags.txt, source lines 20371–20371. Prerequisite pointer invoked at 4701; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0AQ2: sources/stacks-chow/tags.txt, source lines 13495–13495. Prerequisite pointer invoked at 4703; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCL: sources/stacks-restricted/pinned-original-source.tex, source lines 4371–4371. Prerequisite pointer invoked at 4704; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Z \to Y$ be morphisms of locally Noetherian formal algebraic spaces over S. If f is rig-smooth and g is adic, then the base change $X \times_Y Z \to Z$ is rig-smooth.*

Proof. By Formal Spaces, Remark 0GBG and the discussion in Formal Spaces, Section 0AQ2, this follows from Lemma 0GCL. □

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